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A Massively Parallel Basin Simulation with Multiphase Hydrocarbon Migration of a Large-Scale Model

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Abstract

We present a multiphase invasion percolation (IP) feature for hydrocarbon migration recently developed for our massively parallel simulator and showcase the results for a large-scale model. The simulator simulates the generation and migration of compositional hydrocarbon fluids from source rocks to predicted present-day reservoirs, providing a valuable tool for understanding hydrocarbon systems with oil and gas separation. The parallel basin simulator supports exploration teams in decision-making by reducing the uncertainty of the new reservoirs with less simulation times.

Built upon the framework and data structure of a parallel reservoir simulator, our simulator can manage vast, evolving grids in three-dimensional basin models. Geological deposition spanning millions of years is simulated, including compaction, pressure, temperature calculations, and hydrocarbon generation through petroleum kinetics. An in-house Equation of State (EOS) flash routine is part of the simulator. It computes the phase partitioning of hydrocarbon fluids generated from source rocks. Migration of these multiphase fluids is modeled using a massively parallel invasion percolation method and a backfilling algorithm to predict the evolution of oil and gas reservoirs.

Simulation results were verified to ensure consistency with fundamental physical laws and geological principles. We simulated a large geological model ranging stratigraphically from Proterozoic time to the present-day involving 58 different rock types and migration of fluid with four components. A large-scale simulation of about 578 million cells using the multiphase invasion percolation could be completed in 11.32 hours. The model is also simulated at a high grid resolution, comprising over 2.31 billion cells. The results confirm that the multiphase migration is the most computational time-consuming part of the whole-basin simulation. Our contribution lies in developing the multiphase invasion percolation method in a massively parallel basin simulator for simulating the migration of complex hydrocarbon fluids from source rocks to potential reservoirs. This innovation enables rapid and accurate large-scale basin simulations producing multiphase reservoirs, offering significant advantages in terms of speed and reliability for the exploration community.

Introduction

There is an interest of a methodology and a tool that can quantify the risk of a prospect of oil and gas reservoir (charge risk) and help in decision making before the actual drilling. The best way to assess the charge risk is by estimating the presence, types and the volumes of the hydrocarbons in a prospective structure before drilling (Al-Hajeri, et al., 2009). This interest has led to the development of a concept that combines the burial history of a basin with the multiple processes to assess the charge risk in the structures. Basin modelling refers to a study of reconstructing the burial, thermal and maturity history while the petroleum system, Figure 1, refers to the burial, structural and geochemical processes and hydrocarbon generation and its migration in sedimentary structures resulting in the accumulation within the reservoir rock under seals (Magoon & Dow, 1994). Basin and petroleum system modelling (BPSM) is a method used to simulate the geological evolution of a sedimentary basin and the generation, migration, accumulation and preservation of hydrocarbons.

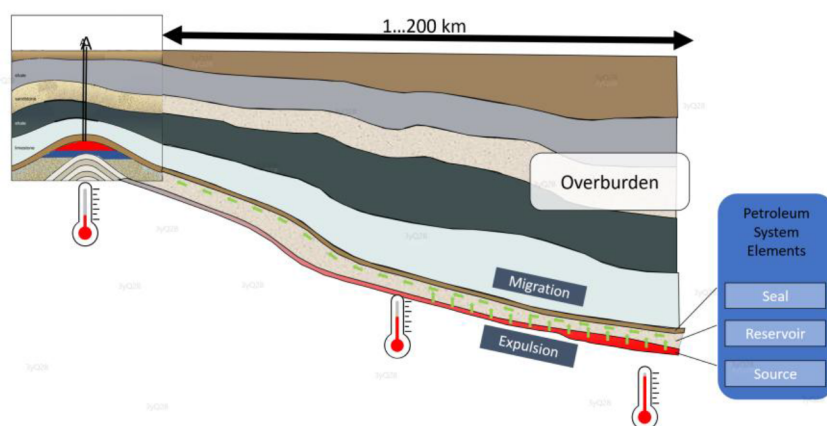


Figure 1—Petroleum system schematic

The first 1D implementation of the BPSM method as a computer simulator was done by Yukler and others at the Nuclear Research Center in Julich, Germany (Yukler, Cornford, & Welte, 1978). The two-dimensional simulation of the groundwater flow was used to estimate rock properties relevant to the migration of hydrocarbons in the software Basin 2 (Bethke, 1985). Later, two-dimensional Darcy's method was the first hydrocarbon migration method implemented in commercial software packages (Ungerer, Burrus, Doligez, & Chenet, 1990) (Hermanrud, 1993). A ray tracing method to simulate the migration of hydrocarbons in highly permeable rocks was also developed (Hantschel, Kauerauf, & Wygrala, Finite-element analysis and ray tracing modeling of petroleum migration, 2000). The combination of a geometrical migration method and Darcy flow was first introduced into 3D basin modelling by Hantschel, et al. (Hantschel & Kauerauf, 2009). Faille, et al. (Faille, et al., 2014) has presented a basin simulator for the 3D simulation of basin models with faults as conduits or barrier to fluid flow.

There has been an interest for simulations of large hydrocarbon basins in Middle East for the geoscientists, explorations and the petroleum engineers. The basin simulator provides the structural properties, abnormal formation pressures, thermal profile and the assessment of the charge risk to help in decision making for exploration of new oil & gas reservoirs. We have developed an in-house basin and petroleum system model simulator (Dogru A. H., et al., 2018), to simulate large-scale basin models with high grid resolution using parallel processing in a high-performance computing system. The simulator is continuously being improved with new functionalities for the users. Here, we briefly present key components of our simulator and the recently developed multiphase invasion percolation method for the compositional hydrocarbon migration. Then, we present simulation results of a large-scale model including the results produced via a recently implemented method of the multiphase invasion percolation for hydrocarbon migration.

Basin Simulator

The in-house basin and petroleum system model simulator, referred to as basin simulator from now on, presented here is designed to simulate large-scale basin models using a highperformance computing (HPC) system. The simulator inherits its framework from the in-house parallel reservoir simulator GigaPOWERS (Dogru A. H., et al., 2009) capable of simulating billions of grid cells over multiple cores of a high-performance computing system in a highly efficient manner. The framework partitions the input grid into multiple simulation grids and distribute these over multiple processors of an HPC system so that computations over all simulation grids are executed in parallel. Additionally, the simulator inherits the parallel linear solvers (Fung & Dogru, Parallel unstructured-solver methods for simulation of complex giant reservoirs, 2008) from the in-house reservoir simulator. These features allow the simulator to turn the simulations of large-scale models around with high speed. Multiple realizations of a large-scale model can be run very fast supporting a more efficient decision-making process in hydrocarbon exploration.

The main input of the basin simulator is stratigraphy data including events of erosion and uplift. Each layer is assigned a deposition age, a present-day thickness and a facies distribution. The facies distribution is linked to a facies data set that includes all the rock types (lithologies) in a model. Source rock characteristics are part of the facies definition containing information on total organic carbon, hydrogen index and kerogen reaction kinetics. The thickness and facies distribution of each geological event in the stratigraphy data are provided in terms of 2D grid maps comprising of uniformly distributed points in X and Y directions specified with a data value. The simulator's input reader can change the resolution of the maps while reading by increasing as well as decreasing the number of points in a map. In this way the overall model grid resolution can easily be upgraded and downgraded as required. Other required inputs are the history of basal heat flow, surface temperature and the paleo-water depths in the form of maps at any specified time of the geological history within the total time period of the stratigraphy data. Once all the input data is read the simulator starts the simulation process comprising of a) Back-stripping b) sequence of processes of each geological event: Figure 2.



Figure 2—A sequence of processes for each time-step of each event of a basin simulation

Components of the simulator

As mentioned above, the simulator is provided with the present-day thickness data corresponding to each event in the stratigraphy data. Back-stripping refers to a method of calculating the original decompacted layer thickness at the time of deposition of each of the deposited sedimentary. Once the simulator has calculated the decompacted thickness of all the layers, it starts the simulation from the oldest time of deposition to the present-day by going through the geological events specified in the stratigraphy data. For each geological event, the simulator calculates the structural, thermodynamic and fluid properties by a sequence of calculations for each time-step of the simulation time as shown in Figure 2: 1) deposition/erosion and compaction for porosity and thickness update 2) pressure calculation 3) heat flow analysis 4) petroleum generation, adsorption and expulsion 5) petroleum migration 6) reservoir volumetrics. Note that porosity, pressure and temperature are computed sequentially at each time-step. For the sake of brevity, we refer to (Hantschel & Kauerauf, 2009) and (Dogru A. H., et al., 2018) for the details of pressure and temperature calculation and the petroleum generation.

Hydrocarbon migration

Once the hydrocarbons are generated by a source rock, expulsion occurs sporadically to release these hydrocarbons. Hydrocarbons are then transported in the carrier rocks in oleic and/or gaseous phases displacing the brine under buoyancy, when the capillary forces are overcome. Solution of the multiphase flow of all phases and components, using the Darcy's law for velocity field, is the most accurate model used in porous media and basin modeling (Hantschel & Kauerauf, Migration and Accumulation, 2009) where time dependency, permeability heterogeneity and the phase relative permeabilities are accounted for. This method is computationally expensive as time-step size is dictated by the numerical stability and extreme heterogeneities. Moreover, basin models are simulated over millions of years of history compared to hundreds of years of simulation as practiced in reservoir simulation. The computational cost increases significantly when large-scale models are discretized into high resolution grids. To circumvent the cost of Darcy's law based multiphase flow method, the invasion percolation method was introduced (Wilkinson, 1984) (Welte, Hantschel, Wygrala, Weissenburger, & Carruthers, 2000). In this method, hydrocarbon movement is primarily driven by the counterbalancing of the buoyancy force and the capillarity between hydrocarbons and water. The flow behavior is assumed to be instantaneous relative to the geological time scale. Here, we use the invasion percolation (IP) with parallel processing (Kayum S., Fung, Port, & Dogru, 2017) (United States Patent No. US11002115B2, 2021) (United States Patent No. US20190228121A1, 2020) for the hydrocarbon migration which is the state of art parallel implementation of the invasion percolation method available in a simulator.

As mentioned above, the water pressure is solved for each time-step of a geological event. We usually solve the hydrocarbon migration at the end of an event (e.g. deposition of a layer) when mobile hydrocarbon is detected using the water pressure field. In the IP method the reservoir is considered to be continuously filled with water. The oil phase potential is computed at each grid cell using the known value of water pressure P_w and the capillary entrance pressure P_{cow} :

$$\Phi_o = P_o - \rho_o g z = P_w + P_{cow} - \rho_o g z$$

For each grid cell i , we compute the difference of oil potential between the neighboring cell j at each side of the cell and its oil potential: $\Phi_{o,ij} = \Phi_{o,i} - \Phi_{o,j}$. The hydrocarbon mass in the grid is transferred to the neighboring cell with which it has the maximum positive potential difference. In this way the mass is transported to the peak of a structure under a seal where it has no further outward movement (no positive potential difference with the neighboring cells). Once the peak cell has accumulated more hydrocarbons than the pore volume other than the critical water saturation, the excess hydrocarbon mass is transferred back to form a trap by a process known as backfilling. The method is illustrated in Figure 3 and Figure 4. As we have an advanced domain partitioning module to sub-divide the grid into sub-domains that are managed by parallel processing in the simulator, the migration is also processed with parallel processing to handle large-scale model quickly. The details of the implementation and challenges of parallel IP, including backfilling, in our simulator have been presented in (Kayum S., Fung, Port, & Dogru, 2017) and (United States Patent No. US11002115B2, 2021).

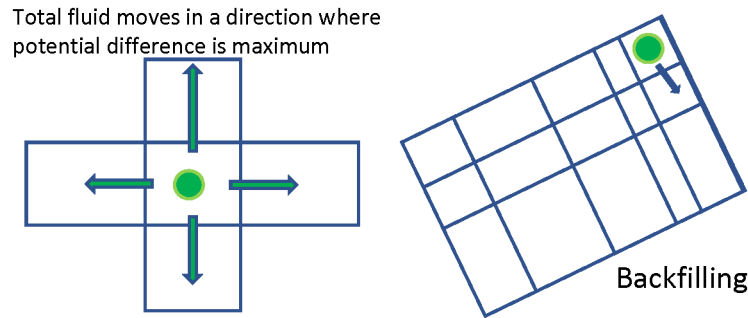


Figure 3—Schematic in vertical cross-section of the hydrocarbon migration and accumulation by invasion percolation method where arrows depict possible directions of flow and green circle depicts oil mass in a cell.

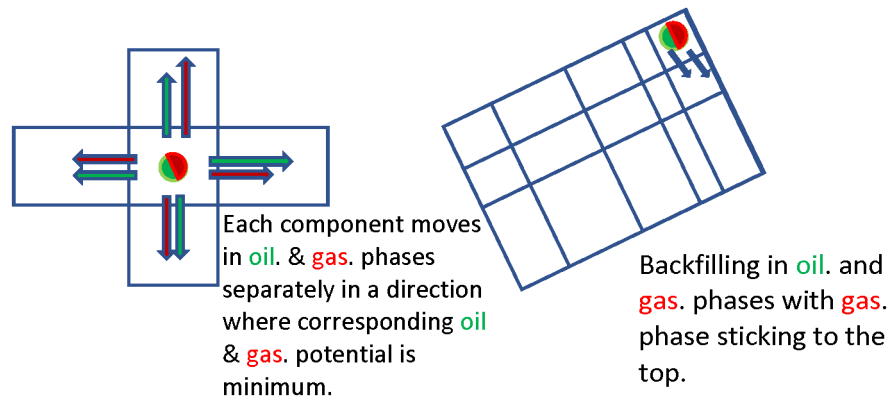


Figure 4—Schematic in vertical cross-section of the hydrocarbon migration and accumulation in two-phases by invasion percolation method where arrows depict possible directions of flow and circle depicts fluid mass in a cell. Red colour illustrates gaseous while green illustrates oil phase.

Multiphase hydrocarbon invasion percolation (IP) migration

Recently, we have also implemented the multi-phase IP method for the migration of the components and backfilling in two phases (oil and gas) instead of migration in a single fluid phase (oil) described above. Once the hydrocarbon components are expelled by the source rock we know the mass of each component c in a grid cell; m_c . We calculate the phase-partitioning of each of the components, at the known temperature and pressure, by the Peng-Robinson equation-of-state (EOS) based flash calculations (an already available in-house implementation). The component mass m_c is composed of two parts then $m_c = x_c m_o + y_c m_g$, where m_o is the total oil mass in the cell and m_g is the total gas mass, x_c is the fraction of oleic mass for component c and y_c is the fraction of the gaseous mass for the component. The total mass in a cell is $m = m_o + m_g = \sum_c m_c$. The flash calculation provides fractions of components (x_c and y_c) for the respective oleic and gaseous phases. In this way, we also obtain the densities of the compositional phases; ρ_o and ρ_g . In addition to the calculation of the oil potential, as mentioned above, we also calculate the gas potential using the gas phase density ρ_g and the gas-water capillary entrance pressure P_{cgw} :

$$\Phi_g = P_g - \rho_g g z = P_w + P_{cgw} - \rho_g g z$$

For each grid cell i , we compute the difference of oil potential as well the gas potential between the neighboring cell j at each side of the cell and its respective oil potential and gas potential: $\Phi_{o,ij} = \Phi_{o,i} - \Phi_{o,j}$ and $\Phi_{g,ij} = \Phi_{g,i} - \Phi_{g,j}$. We follow the invasion percolation method to transfer the mass of each component in oleic phase ($x_c m_o$) and gaseous phase ($y_c m_g$) in a cell to a direction dictated by the respective maximum positive difference in oil and gas potentials, illustrated in Figure 4. If there is a neighbor oil trap then the migrated oil is added to that oil trap while, the gas phase is allowed to approach the peak of a structure forming a gas cap or combine with an already forming gas cap above oil trap. As mentioned before, the

entire model is partitioned into sub-domains for parallel processing and migration is also simulated for all the sub-domains in parallel. The fractions of the components and densities of the compositional phases are updated via flash calculation after each migration step as described above.

Reservoir volumetric

As the migration of hydrocarbon progresses and peaks of the structures are filled with excess hydrocarbon mass, the backfilling mechanism is processed with an innovative parallel implementation as mentioned above. The possible trap peaks are first identified as a first step of the migration processing once the hydrocarbon mass is released by the source rocks. The excess hydrocarbon mass (more than the possible mass determined by the pore volume) in a trap peak is transferred back to a grid cell with minimum potential among all the boundary cells (neighboring cells for the first cell at the peak where trap formation is started) to form a trap which is known as backfilling. The new cell receiving the excess mass is designated as a trap tip. Note that initially all peaks are also designated as trap tips when no backfilling has started. This step of backfilling is continued until no excess mass is left in the trap forming cells or the trap is merged with another trap. The backfilling processing on parallel processors is challenging to implement as a trap can grow into multiple sub-domains managed by multiple processors (Kayum S., Fung, Port, & Dogru, 2017).

In the multiphase hydrocarbon case, we process the backfilling of the oil mass in the trap to form an oil trap and then process the backfilling of gas mass to form a gas cap above an oil trap. If the total volume of the multiphase fluid at a peak is more than the pore volume, the oil mass, determined by the excess volume, is backfilled to form an oil trap and leaving the gas mass at the trap-peak (peak of the structure). In the next step any gas mass in the excess volume is also backfilled to form the gas cap. The mass of each component out of the excess oleic mass ($x_o m_o$) in a cell at the trap peak is transferred to a grid cell in the trap boundary with minimum oil potential. In the same way, the mass of each component out of the excess gaseous mass ($x_g m_o$) in a cell at the trap peak is transferred to a grid cell in the trap boundary with minimum gas potential. Figure 3. illustrates the flow of the multiphase hydrocarbon migration and accumulation process.

Once the 3D traps are determined, the simulator calculates the 2D concave polygons, each bounding one of the traps and writes in a binary format using the shape format (ESRI Shapefile Technical Description, 1998) known as shapefiles. We also calculate and report the compositions, hydrocarbon masses, volumes and minimum and maximum depths of the traps in terms of the attributes of the shapefiles.

Synthetic test model

We have built a synthetic layer-cake model for testing purposes with five layers representing the underburden, a source layer, reservoir, seal and overburden as shown in Figure 5. The reservoir has a structure that is expected to be filled with hydrocarbons. The grid has discretization of 200 cells both X and Y directions. The source rock uses a Type-II(B) kinetic reaction (Baur, 2019). We simulate the model for a geological history of 50 million years with a uniform basal heat flow of 70 mW/m². The hydrocarbon migration and accumulation are computed using the single fluid phase IP (default method) as well as the multiphase IP implementation. Figure 6. shows the hydrocarbon accumulation predicted for the present-day simulated by the single fluid phase IP method (default method) as well as the new multiphase IP method presented here. The single phase IP method consumes 2.13 minutes of CPU time using 128 processors (cores) while multiphase IP method consumes 3.06 minutes as it involves flash calculations during migration and accumulation processing of gas and oil phases separately in the traps by backfilling. The overall hydrocarbon accumulations are comparable for both the methods. Figure 7 presents hydrocarbon accumulations predicted by a commercial simulator. The results are comparable to our simulator. The differences in the hydrocarbon volumes can be explained by the application of different discretization methods (finite-element vs finite-volume) affecting the temperature solution. Other differences cannot be inferred as the numerical details of the commercial simulator are not publicly available.

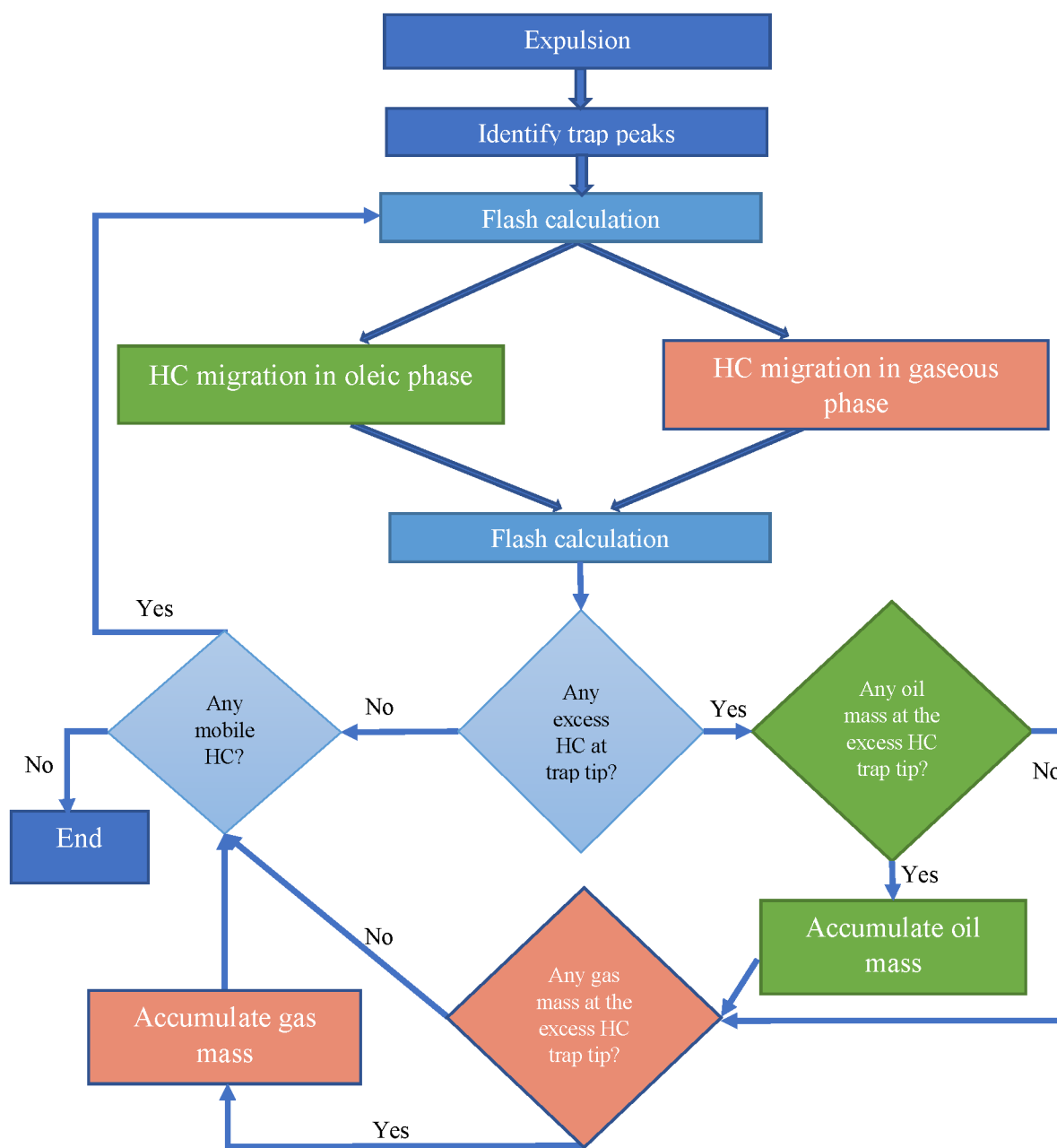


Figure 5—Flow diagram illustrating the multiphase hydrocarbon migration and accumulation method according to an implementation

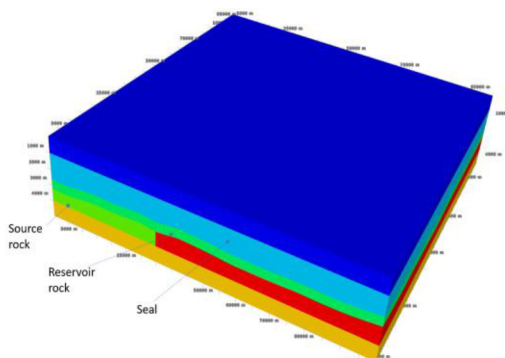


Figure 6—Synthetic layer-cake model for testing

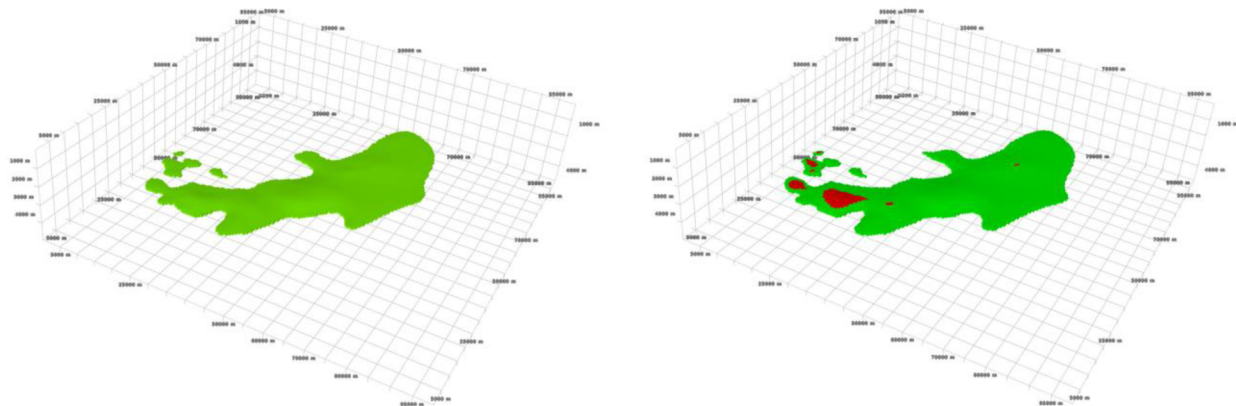


Figure 7—Left: Hydrocarbon accumulation at present-day of a synthetic layer-cake model predicted by single-phase IP method of the simulator. Right: Hydrocarbon accumulation estimated by the multiphase IP with red color depicting gas phase.

Large scale model simulation

Here, we demonstrate the capability of our simulator by running a simulation of a large-scale basin model. Figure 8 shows the outline of a basin model. The input data is obtained from the public domain database of Neftekh (Neftekh Predictions geological framework, 2025). We have a total of 296 layers with the first one started deposition at 632.5 million years. There are 58 different rock facies types including shale and sand stone. The model is lustrated in Figure 9 where vertical scale is exaggerated for visualization purpose. Each layer of the model is discretized into uniformly distributed cartesian cells with size of 1000. meters in X as well in Y direction, resulting in 1,896 cells in X direction and 1,031 cells in the Y direction. The total number of cells in the whole model is then equal to $1,896 \times 1,031 \times 296 = 578,613,696$ (578.6 million). We also have two layers that include the source rocks as one of the layers is shown in Figure 10. Kinetic reaction model is the Type II(B) of (Baur, 2019), specified for the source rock, which produces four hydrocarbon components. History and distribution of surface temperature, basal heat flow and paleo-water depth are also provided.

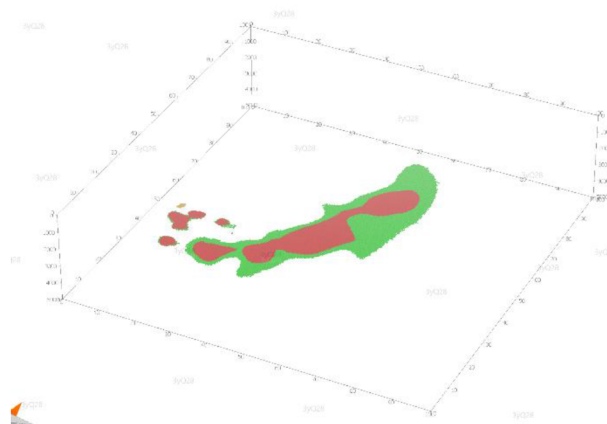


Figure 8—Hydrocarbon accumulation at present-day of the synthetic layer-cake model as predicted by a commercial simulator.

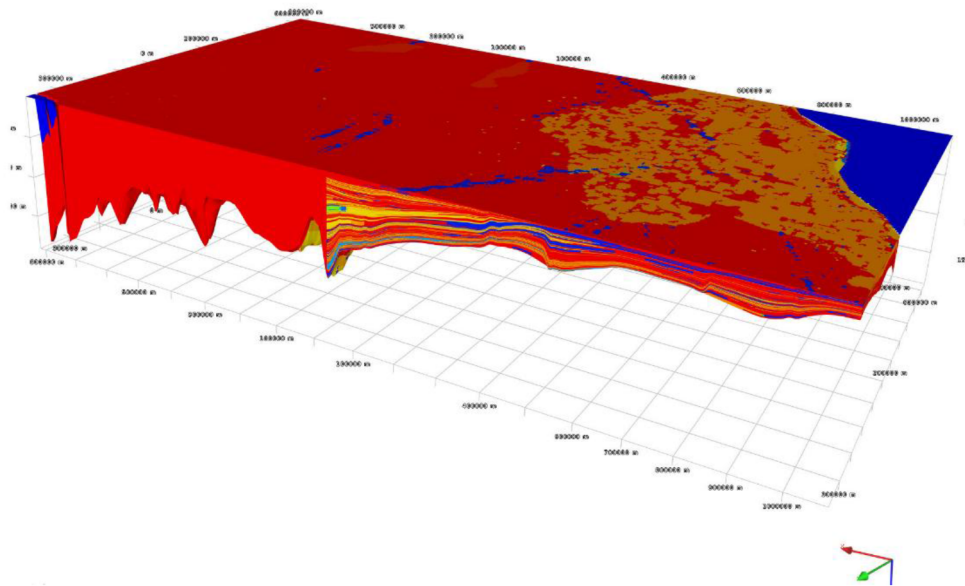


Figure 9—Top view of the outline of the large basin model with different colors depicting different facies

We have simulated the model using 4096 cores (64 nodes) using our company advanced HPC system Ghawar-1 ([Top 500 The List., n.d.](#)). Migration of hydrocarbon is simulated by the single fluid phase IP method for 632.5 million years of geological history. The total run time of the simulation is 4.97 hours. The CPU time consumed by hydrocarbon migration and accumulation is around 55% of the total simulation time of this model. In our experience migration can consume up to 85% of the total time when many more source rocks are included in a model resulting in thousands of traps. We also simulated the model with grid-refinement by dividing each grid-cell in a layer into four cells resulting in a model of $3,792 \times 2,062 \times 296 = 2,314,454,784$ (2.31 billion). The total simulation time of this fine grid model using different number of cores are reported in [Table 1](#) which shows the decrease in computational time with the increase in number of cores.

Table 1—Total run time of the simulations of a large model with grid-refinement via single-phase (default) IP method using different number of cores

Number of cores / Number of nodes	Simulation time in hours
8192 /128	19.97
16384 /256	15.467
32768 /512	13.476

The temperature profile of a layer with source rock properties and its transformation ratio are shown in [Figure 10](#). A cross-section of the model with the temperature distribution is also shown in [Figure 11](#). As hydrocarbon is generated by the source rock, it is expelled into the overlaying carrier rocks. Hydrocarbons migrate via the carrier rocks towards the structural peaks under seals where these are accumulated to form traps. The traps at present-day as predicted by the default single fluid phase IP method with parallel processing are shown in [Figure 12](#). The production of the thermodynamic results and prediction of the traps within a work day demonstrates the capability of our simulator and HPC system to turn models around quickly.

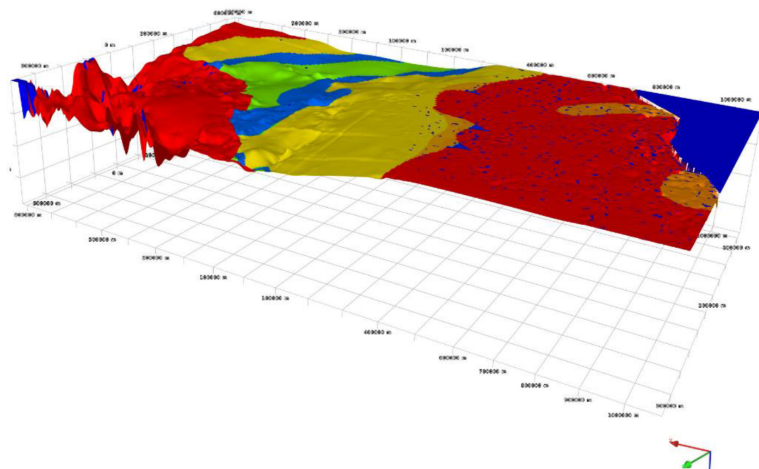


Figure 10—Facies of a layer with a source rock (green).

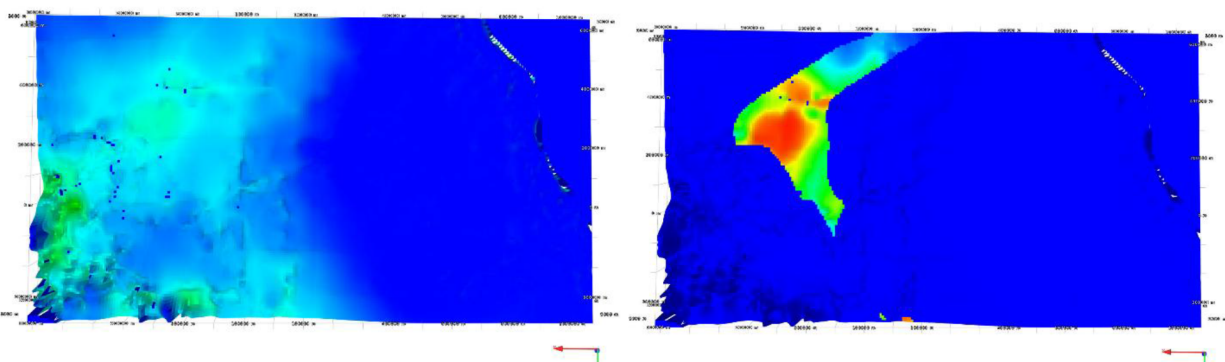


Figure 11—Temperature profile and transformation ratio, at present-day, of a layer with the source rock.

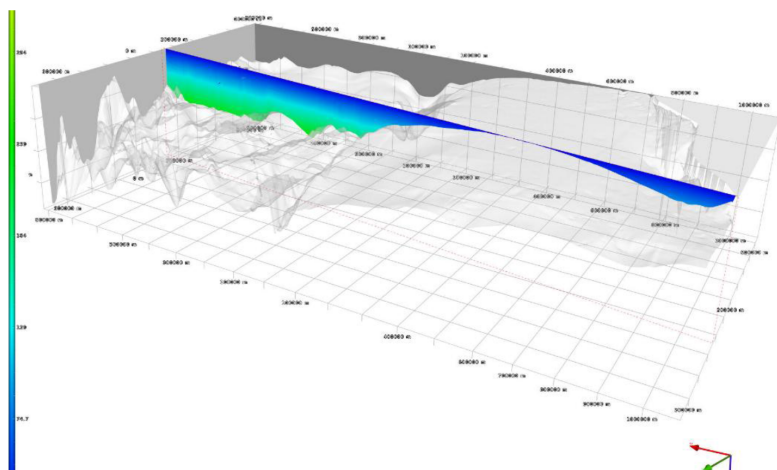


Figure 12—Cross-sectional view of the temperature of the large basin model.

Referring to the section (hydrocarbon migration) regarding a new implementation of multiphase invasion percolation (IP) method with flash calculations during migration, we also simulate the model with the option using 4096 cores (64 nodes). The total run-time reported is 11.32 hours which is higher than the default single fluid phase IP method because of flash calculations throughout the migration process and the processing of oil and gas traps with distinction instead of traps with mixed components in single phase. The traps at present-day as predicted by the multi phase IP method with parallel processing are shown in Figure 13. Gas traps are depicted in red color while oil traps are depicted in green color. Figure 14 shows the 2D view of

the trap polygons produced by the default method and the new implementation which shows the difference in the prediction of gas traps. The total simulation time of the fine grid version of the model using different number of cores are reported in [Table 2](#).

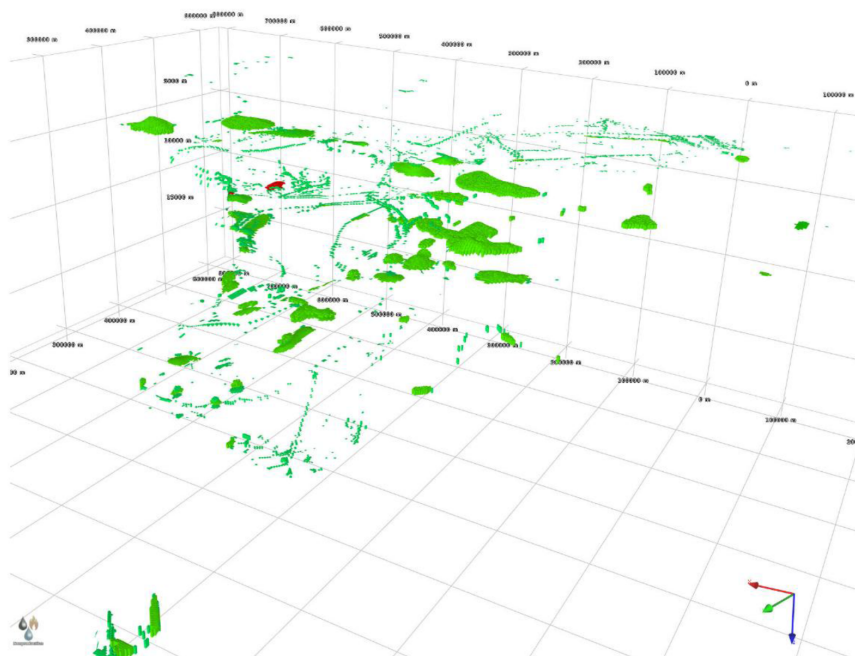


Figure 13—The hydrocarbon accumulations (traps) as predicted by the simulator in a 3D view by single-phase IP method.

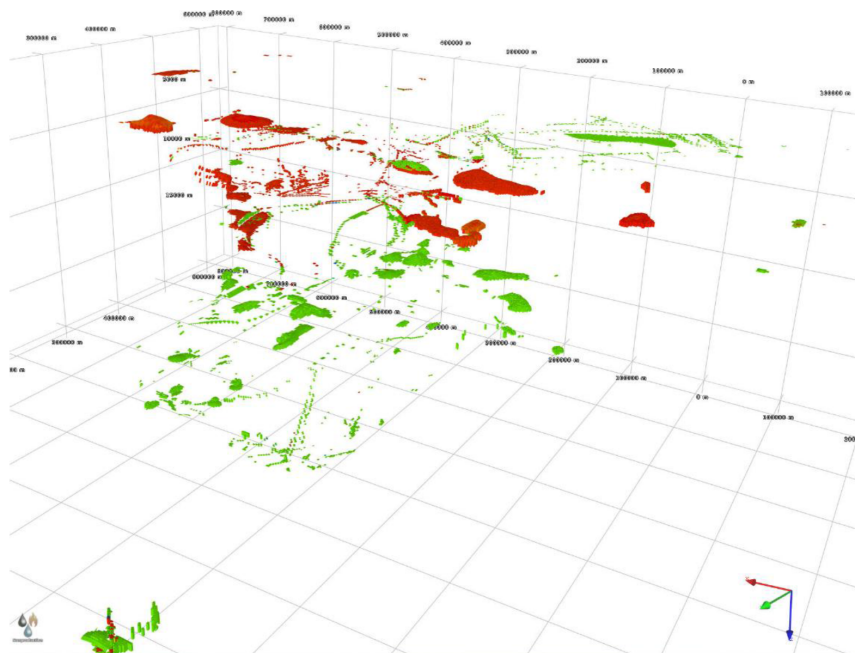


Figure 14—The hydrocarbon accumulations (traps) as predicted by the simulator in a 3D view by multiphase IP method

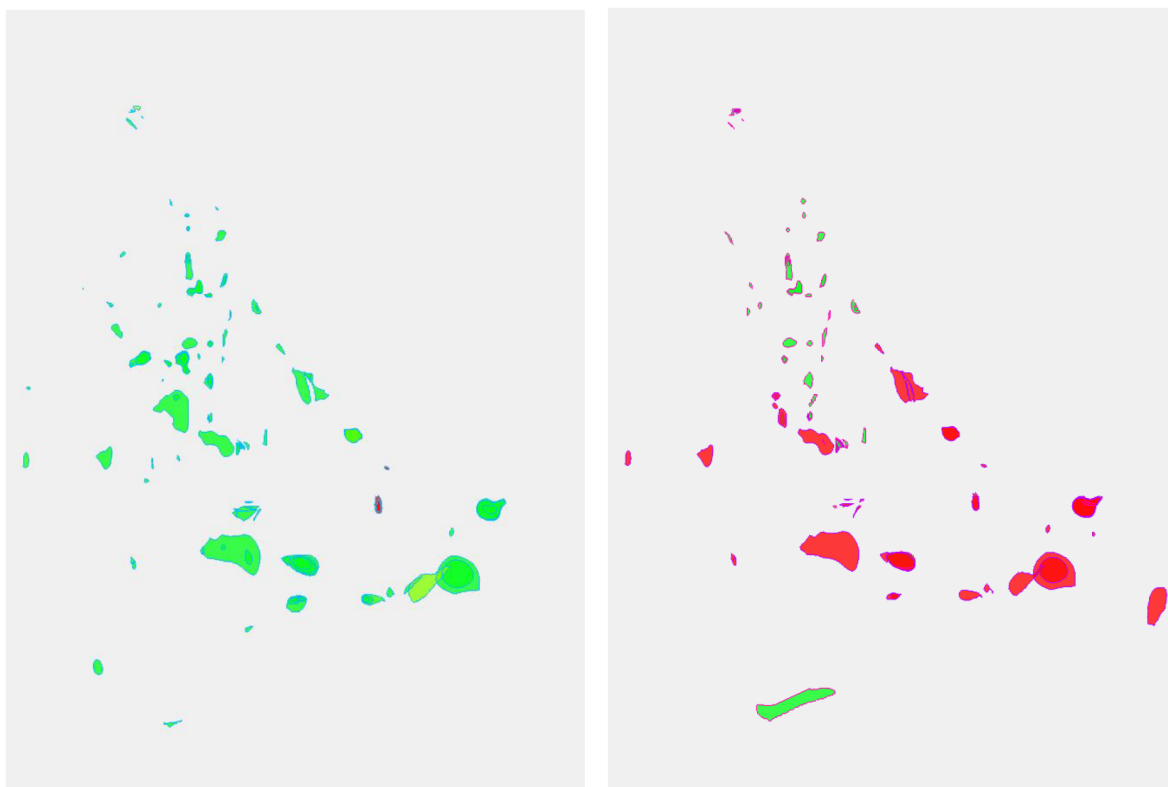


Figure 15—2D view of traps polygons as produced by default IP method (left) and the multiphase IP migration method (right).

Table 2—Total run time of the simulations of a large model with grid-refinement via multiphase IP method using different number of cores

Number of cores / Number of nodes	Simulation time in hours
16384 /256	21.24
32768 /512	18.91

We propose to use the default single fluid phase IP method for simulations for multiple scenarios and run the simulation using the multiphase IP method for the calibrated model whenever a distinction the oil and gas phases is required.

Conclusions

We present a large-scale basin simulation including multiphase hydrocarbon migration using massively parallel processing. The simulator employs the advanced domain-partitioning tool and data structure and deals with the large-scale grid with multiple processors efficiently. In addition to the already implemented single-phase invasion percolation (IP), we have implemented the multiphase invasion percolation method with parallel processing for hydrocarbon migration and accumulation processing at each geological event. A synthetic model is simulated which shows consistent result produced by the simulator. Simulation results of a large-scale model using the multiphase IP is presented which demonstrates the capability of our massively parallel simulator to predict oil and gas traps. The fine-grid model is simulated using different number of cores, showing the decrease in computational time with the increase in the number of cores.

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